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Nurses' Perceptions of Their Practice Environment in Public Hospitals: A Repeated Cross-Sectional Study

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ABSTRACT

The study aims to investigate registered nurses' perceptions of their practice environment in public hospitals. A 12-month repeated cross-sectional study with quarterly surveys was conducted in 71 adult medical and surgical wards in four public hospitals in Hong Kong. The perceptions of nurses' practice environment were measured by the 31-item Practice Environment Scale of the Nursing Work Index administered in four quarterly survey waves (ranging from 604 to 865 participants). Four of the five subscales scored above 2.5 on average, indicating that these wards were considered favorable practice environments. The only subscale rated unfavorably was "Staffing and resource adequacy."

KEYWORDS

Nurse practice environment; nursing education; quality of care; staffing; workload

Introduction

The work environment is fundamental to employee efficiency and their ability to successfully meet work goals. In the healthcare context, nurses' ability to deliver high quality patient care is similarly affected by aspects of their practice environment, including management, nurse leadership, sufficiency of staff and resources, collegiality and collaboration, and involvement in hospital affairs (Kieft et al. 2014,Kim, Yoo, and Seo 2018). A recent meta-analysis of 17 studies from over 22 countries found that positively-rated work environments were associated with lower odds of negative patient outcomes and higher odds of patient satisfaction, indicating the benefits of a favorable setting for nursing practice (Lake et al. 2019). However, as nurse turnover rates continue to rise globally and staff burnout persists as a major concern, particularly with the impact of the recent COVID-19 pandemic (Xu, Intrator, and Bowblis 2020), working conditions have become more challenging and there is often inadequate structural or material support to assist nurses in maintaining a high standard of care while managing growing workloads.

Similar to the global context, Hong Kong faces a serious shortage of nurses which has been aggravated by the COVID-19 pandemic as well as persistent concerns regarding suboptimal work conditions in healthcare settings. In a recent qualitative study of nurses who had resigned from public hospitals in Hong Kong and switched occupations, turnover was found to be highly influenced by workplace factors such as a tense working environment, insufficient opportunities for professional development, and poor communication with supervisors and hospital leadership (Hung and Lam 2020). Previous studies also suggest that nurses experience more depression and anxiety compared to the general public and that

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intention to quit was influenced by job stress (Cheung and Yip 2015). Moreover, there is also evidence that newly-graduated nurses face challenges in speaking up and advocating for patient safety due to a poor culture of communication which discourages nurses from openly discussing their concerns (Law and Chan 2015). As such, to empower nurses in ensuring the integrity and quality of healthcare services, it is necessary to address nurses' concerns about their working conditions in order to bring improvements to their practice environment, boost nurse retention, and facilitate the delivery of high-quality care.

The Practice Environment Scale of the Nursing Work Index (PES-NWI) is a widely used instrument that has been adopted globally to measure nurses' perceptions of their working environment (Swiger et al. 2017). The scale is frequently used to identify associations between the practice environment and nurse outcomes such as resilience, job satisfaction, and intention to leave (Tabakakis et al. 2019). However, the PES-NWI has not been used to evaluate nurse's perceptions of their practice environments in Hong Kong. This study was therefore conducted to address a key research gap and develop a deeper understanding of nurses' experiences of their workplace and, where possible, provide evidence that can be used to optimize nurses' practice and patient care.

Methods

Study Design and Setting

A repeated cross-sectional study was conducted from January 2016 to December 2018 in four public hospitals that offer 24-h emergency services (out of 18 public hospitals offering emergency services in Hong Kong) to investigate nurses' perceptions of their practice environment. Seventy-one adult medical and surgical wards (including general surgery and orthopedic wards) were recruited. This study was conducted as part of a larger research project investigating associations between nurse staffing, nurse practice environment and nurse-sensitive patient outcomes, the protocol of which has been published elsewhere (Chau et al. 2015).

Sample

Data were collected from all full-time registered nurses (RNs) working in the study wards with direct patient care duties for 50% or more of their job responsibilities. Direct patient care duties are defined as patient monitoring and examination, medication administration, health education and communication, and care coordination and planning.

Measures

Nurse Practice Environment

The PES-NWI is a reliable instrument to evaluate features of nurses' working environment and the scale has been widely adapted for global use (Swiger et al. 2017, Lake 2002). In this study, the Chinese version of the PES-NWI, which has been validated and found to have reliable psychometric properties (You et al. 2012), was applied. The scale comprises 31 items divided into the following five subscales; (1) Nurse participation in hospital affairs; (2) Nursing foundations for quality of care; (3) Nurse manager ability, leadership, and support of nurses; (4) Staffing and resource adequacy, and; (5) Collegial nurse-physician relations. Each item is scored from 1 (Strongly disagree) to 4 (Strongly agree), with a higher score indicating more positive evaluation. Composite and subscale scores are derived from averaging the relevant items. A composite score above 2.5 indicates a favorable practice environment, whereas a score of 2.5 or below indicates an unfavorable environment. To better illustrate the overall quality of individual nursing practice environments while accounting for scoring variations among different subscales, Lake and Friese proposed a three-category classification system where environments were considered favorable if they scored above 2.5 in four or five subscales, mixed if they scored above 2.5 in two to three subscales, and unfavorable if they scored above 2.5 in one or no subscale (Lake and Friese 2006).

Sociodemographic Information

Sociodemographic data including sex, age, education level at entry and at present, current job position, and years of experience were collected.

Data Collection Procedures

The project team maintained regular correspondence with the participating hospitals and received approval to carry out data collection in the participating wards. Hard copies of the PES-NWI and sociodemographic surveys were distributed to RNs in participating units by a research assistant from the project team. A collection box was set up in each hospital to facilitate participants in returning the completed surveys. In order to achieve a more comprehensive understanding of nurses' practice environment and avoid some seasonal influences, such as peak illness period due to seasonal influenza, surveys were administered every three months over a 12-month period.

Statistical Analyses

Data were summarized and presented using appropriate descriptive statistics. The quarterly survey data were presented separately and aggregately to provide an overview of the outcome measures over the 12-month study period. Specifically, the mean subscale scores of the PES-NWI were used to quantify the domains of practice environment. The scale reliability of the PES-NWI was assessed by using Cronbach's alpha coefficient based on the entirety of survey data accumulated over 12 months. All statistical analyses were performed using IBM SPSS 26.0 (IBM Crop., Armonk, NY)

Ethical Approval

Ethical approval was received from the Joint CUHK-New Territories East Cluster Clinical Research Ethics Committee and the Kowloon West Cluster Research Ethics Committee (CRE-2013.101). Eligible participants were provided an explanation of the study purpose and informed written consent was obtained. Participants' rights and safety were safeguarded in accordance with local laws, the Declaration of Helsinki, institutional policies, and the ICH-GCP. All questionnaires and data analysis were conducted anonymously.

Results

Sociodemographics

The number of participants in each of the four survey waves ranged from 604 to 865. Over 70% of participants were female and aged 40 or below. Registered nurses comprised the largest group of participants (>80%), followed by advanced practice nurses (>10%). Most participants (>70%) spent 76–100% of their time on direct patient care on average. Approximately half of the participants had attained Bachelor's or Master's degrees prior to registration. Participants had an average of 5.4 years of experience as an RN and 3.9 years of experience in their current position. Participants' average work duration in their current ward ranged from 28 to 36 months. The summary of participant characteristics can be viewed in Table 1.

Nurse Practice Environment

Participants' perceptions of their practice environment as evaluated through the PES-NWI and the scale's reliability coefficients are reported in Table 2. Cronbach's alpha coefficients of the PES-NWI subscales and overall scale ranged from 0.72 to 0.93, which supports the reliability of the scale. The average composite score across the four waves of survey was over 2.6, indicating that the study units were considered to be favorable practice environments on the whole. Four out of five subscales had average scores >2.5, of which the highest scoring subscales were "Nursing foundations for quality of care", "Nurse manager ability, leadership, and support of nurses", and "Collegial nurse-physician relations", with scores reaching 2.8. The subscale of "Nurse participation in hospital affairs" also achieved a favorable average score of >2.6, with the exception of two items "Opportunity for staff nurses to participate in policy decisions" and "Staff nurses are involved in the internal governance of the hospital", which scored 2.5 or below.

In terms of the highest scored individual items, two items under the subscale of "Nursing foundations for quality of care" had a mean score of 3.0 or above across the four surveys. The items

Table 1. Sociodemographic characteristics of the nurse participants in the surveys.

	Wave of survey			
Characteristics	First (n = 865)	Second (n = 604)	Third (n = 680)	Fourth (n = 661)
Sex				
Female	74.3%	74.4%	78.2%	75.9%
Male	25.7%	25.6%	21.8%	24.1%
Age (years)				
≤25	21.1%	22.5%	19.9%	22.8%
26–30	31.5%	34.7%	35.8%	37.1%
31–35	11.9%	10.6%	10.5%	10.4%
36–40	10.1%	10.9%	12.7%	10.4%
41–45	6.9%	8.1%	9.1%	8.1%
46–50	5.6%	7.6%	6.9%	6.8%
51–55	3.9%	4.0%	4.2%	3.2%
56–60	1.1%	1.6%	0.9%	1.0%
>60	7.9%	0.0%	0.0%	0.3%
Current job title				
RN	82.0%	83.3%	82.3%	82.6%
APN	13.0%	12.4%	13.7%	13.8%
NO	1.4%	1.8%	1.3%	1.0%
NS	0.4%	0.7%	0.6%	0.7%
WM	3.1%	1.8%	1.9%	1.8%
NC	0.0%	0.0%	0.2%	0.2%
Average working time spent on direct patient care				
51–75%	30.0%	29.3%	29.3%	29.5%
76–100%	70.0%	70.7%	70.7%	70.5%
Educational level at entry of RN				
Diploma	49.1%	50.5%	50.0%	47.3%
Bachelor pre-registration	47.9%	46.5%	46.7%	50.9%
Master pre-registration	3.0%	3.0%	3.3%	1.8%
Highest educational attainment				
Diploma	13.2%	14.5%	12.2%	13.4%
Bachelor	60.2%	59.9%	61.4%	60.1%
Master	25.9%	24.8%	25.0%	24.6%
DN	0.0%	0.0%	0.6%	1.3%
PhD	0.0%	0.0%	0.2%	0.0%
FHKAN	0.7%	0.9%	0.6%	0.7%
Years of experience in current position†	4.0 (2.0–8.3)	4.1 (1.7–9.0)	3.9 (1.9–8.0)	3.7 (1.9–7.4)
Years of experience as registered nurse†	5.7 (2.4–14.0)	5.0 (2.0–14.3)	5.4 (2.1–15.0)	5.3 (2.3–14.7)
Working duration in current ward (months)†	36 (12–84)	28 (9–79)	33 (12–84)	30 (12–85)

Data of variables marked with† are presented as median (inter-quartile range), all others are presented as %.
Abbreviations: APN: advanced practice nurse; DN: Doctor of Nursing; FHKAN: Fellow of Hong Kong Academy of Nursing; NC: nurse consultant; NO: nursing officer; PhD: Doctor of Philosophy; RN: registered nurse; NS: nurse specialist: WM: ward manager.

Table 2. Perceived practice environment by the nurse participants.

	Scale reliability	Wave of survey				Average
Practice Environment Scale of the Nursing Work Index (PES-NWI)	Cronbach's α	First	Second	Third	Fourth	Over the year
Nurse participation in hospital affairs	0.85	2.62 (0.40)	2.64 (0.40)	2.63 (0.41)	2.64 (0.43)	2.63 (0.41)
Nursing foundations for quality of care	0.80	2.81 (0.31)	2.80 (0.31)	2.80 (0.32)	2.80 (0.33)	2.80 (0.32)
Nurse manager ability, leadership, & support of nurses	0.80	2.81 (0.43)	2.81 (0.43)	2.79 (0.43)	2.80 (0.45)	2.80 (0.44)
Staffing and resource adequacy	0.83	2.10 (0.54)	2.10 (0.54)	2.17 (0.54)	2.20 (0.57)	2.14 (0.55)
Collegial nurse-physician relations	0.72	2.81 (0.39)	2.79 (0.39)	2.77 (0.42)	2.77 (0.40)	2.79 (0.40)
Composite score	0.93	2.63 (0.33)	2.63 (0.33)	2.63 (0.34)	2.64 (0.35)	2.63 (0.33)

Data are presented as mean (standard deviation).
The possible ranges of the five domain and composite scores of the PES-NWI all range from 1 to 4.

are "High standards of nursing care are expected by the administration" and "A preceptor program for newly hired RNs".

The lowest scored subscale was "Staffing and resource adequacy", which scored below 2.2 on average. In particular, two items about staffing adequacy ("Enough registered nurses to provide quality patient care" and "Enough staff to get work done") received scores below 2, the lowest scores in the scale.

Regarding differences in nurses' perceptions over the 12-month period, no significant changes in scores were observed, reflecting no major modifications to the environment during the study

period. However, scores of the "Staffing and resource adequacy" subscale improved steadily from 2.1 to 2.2 across the four survey waves.

Discussion

This is the first study to explore registered nurses' perceptions of their practice environment in major hospitals in Hong Kong and our findings are largely in line with prior international research, suggesting that nurses in the present study face similar work concerns as their global counterparts.

In particular, issues with staffing emerged as the aspect needing most improvement as the "Staffing and resource adequacy" subscale had the lowest scoring items and was the only category rated unfavorably on the scale. This is a common finding worldwide, as several studies show that staffing is a forefront concern for most healthcare institutions and even in magnet-certified facilities (Swiger et al. 2017, Bruyneel et al. 2023). As direct patient care duties often comprise nurses' primary responsibility, nursing shortages significantly reduce the time that nurses are able to dedicate to each individual patient, thus resulting in inadequate care provision and low patient satisfaction. Missed care for example, is a common occurrence in short-staffed units which both adds to nurses' work stress and leads to poor patient outcomes (Missed Care Study Group, 2018). There is thus an urgent need to reevaluate nurse recruitment and retention policies in order to reduce turnover, maintain an appropriately-sized workforce, and facilitate nurses in providing a high standard of care. Indeed, a recent longitudinal study with data collected from 458 US hospitals over a 10-year period found that improvements in both working conditions and nurse staffing resulted in decreased likelihood of missed care (Lake, Riman, and Sloane 2020), demonstrating that enhancements to nurses' work environments are a worthwhile investment for patient benefit. There is also promising evidence indicating that educational interventions may support nurse retention, with resilience training and mentorship programmes for newly graduated nurses resulting in improved stress management and lower nurse turnover (Zhang et al. 2019, Chesak et al. 2021).

Besides staffing, nurses' level of involvement and participation in hospital affairs emerged as another area that requires attention. Though the subscale was rated favorably on the whole, two items regarding opportunities for nurses to participate in policy decisions, and involvement in internal hospital governance scored unfavorably, indicating nurses' unmet demand for representation in upper-level management and insufficient chances to contribute to policy decision-making. Despite growing recognition and emphasis on the value of nurses' involvement and contribution to health policies in recent years (Turale and Kunaviktikul 2019), our findings are consistent with international evidence suggesting that nurse participation in the development of health policy and hospital affairs remains suboptimal (Kunaviktikul et al. 2010; AbuAlRub and Abdulnabi 2020; Rasheed, Younas, and Mehdi 2020). Several barriers have been identified, including inadequate knowledge of health policy itself, a lack of time and resources to enable participation, as well as macro factors such as gender disparities in senior management and hospital boards, and poor societal perceptions of the nursing profession which lead to undervaluation of nurse perspectives (Hajizadeh et al. 2021; Prybil 2016). However, there have yet to be major improvements in terms of organizational, structural, or cultural changes to empower nurses in sharing their professional expertise and raising their concerns at the institutional and policy level. Moreover, in Hong Kong particularly, substantial gaps remain in the evidence on nurses' experiences and involvement with policy-making and hospital governance which creates a challenge for stakeholders in devising effective plans to address the barriers faced by nurses. Considering that nurse input can greatly benefit hospital and healthcare policies, further research needs to be conducted on methods to increase nurse involvement and examine its effects on nurse outcomes such as turnover, and patient outcomes so as to improve nurses' sense of participation within their workplace.

On a positive note, the remaining subscales were all rated favorably with no items receiving a score below 2.5. Notably, the subscale of "Nursing foundations for quality of care" had

the highest-scoring items with two items receiving an average score above 3, suggesting that nurses had a good understanding of the quality demanded for patient care and felt empowered to provide that through their wards' established strategies and processes. Nurse managers' support and leadership abilities were also evaluated favorably, presenting an encouraging finding as characteristics such as leadership style have been associated with outcomes such as job satisfaction and nurse-reported adverse events including patient complaints and medication errors (Labrague 2021; Morsiani, Bagnasco, and Sasso 2017). In terms of collegial nurse-physician relations, the current findings are again reassuring as previous evidence suggests that nurses who perceive poor relationships with physicians are more likely to report unfavorable patient outcomes (Mihdawi et al. 2020). As negative interpersonal relationships at the workplace have been found to reduce nurses' practice quality, resulting in ineffective teamwork and impaired critical thinking skills (Sahay and Willis 2022), the finding that nurses in the present study rated their ward leadership and collegial relations favorably is a welcome one.

However, considering the growing concern of nursing shortages which continues to generate significant work stress for existing staff and nurses, besides improving staffing policies, more research into nurses' work relationships and how they could aid in mitigating the harmful impacts of work stress would be useful in devising strategies to facilitate collegiality, enhance teamwork, and improve nurses' perceptions of the practice environments on the whole. In addition to reevaluating workplace policies, educational strategies to better prepare and support nurses in managing workplace challenges may also be a viable approach to optimizing nurses' experience of the work environment.

Limitations

Our study has some limitations. Although the surveys were conducted in all 71 wards of four major hospitals, as only adult medical and surgical wards were involved and the sample may be subject to self-selection bias, the findings may not be generalizable to other settings and reflect the overall nurse practice environments in Hong Kong. Furthermore, the 12-month follow-up period has resulted in stable PES-NWI scores across the quarterly surveys, suggesting that no major transformations took place within participants' wards during the study period. A longer follow-up period may have been able to capture any potential changes in the practice environment over time and allowed for more investigation into nurses' perceptions regarding the changes. Moreover, there is a lack of turnover data from each unit which could have supplemented findings on staff inadequacy in the participating hospitals.

Conclusion

Findings in the present study demonstrate that nurse practice environments are generally assessed favorably by nurses and that results across the subscales are in line with previous international studies. However, there is particular room for improvement in terms of ensuring adequate staffing and boosting nurse involvement in hospital affairs. More efforts must be dedicated to encourage nurse retention and staff development through appropriate interventions such as relevant training programmes, and organisation-level changes must be made to enable and increase nurse participation, engagement and influence in hospital governance.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author, SHSL, upon reasonable request.

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